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Assignment -3
Neural Network and Deep Learning

Q1. Numerical: Compute weight updates using backpropagation.
Ans.
Given (example):

Input $x=1$, weight $w=0.5$, bias $b=0$
Target $y=1$, learning rate $\eta=0.1$
Activation = sigmoid

Step 1: Forward pass

$z=wx+b=0.5$

$a=\sigma(z)=1/1+e^{-0.5} = 0.62$

Step 2:

Error

$E = 1/2(y - a)^2$

Step 3: Gradient

$dw/dE = (a - y) \cdot a(1 - a) \cdot x$
$= (0.62 - 1)(0.62)(0.38)(1) = -0.089$

Step 4: Weight update
$wnew = w - \eta$
$dw/dE = 0.5 - (0.1)(-0.089) = 0.5089$

Q2 Numerical: Apply L2 regularization to a given loss function.
Ans.
Loss with L2:
$L = L0 + (\lambda/2)\, w^2$

Example:
$L0 = 2,\ w = 3,\ \lambda = 0.1$
$L = 2 + (0.1/2)(3^2)$
$= 2 + 0.05 \times 9$
$= 2 + 0.45 = 2.45$

Effect on update:

$w = w - \eta (dL0/dw + \lambda w)$

Q3 Explain autoencoders and their use in feature extraction.

Ans. Neural networks that learn to reconstruct input

Structure:

Encoder $\longrightarrow$ compress data

Decoder $\longrightarrow$ reconstruct data

Use:

- Dimensionality reduction
- Feature extraction
- Data compression

Q4 Case Study: Analyze the use of autoencoders in noise reduction.

Ans. Use Denoising Autoencoder

Input = noisy data

Output = clean data

Working:

Model learns to remove noise during

training.
Extracts robust features.
Applications:
- Image denoising
- Speech enhancement

Q5 Explain the terms:
(i):- deterministic and probabilistic neural networks
- Deterministic:
Same input ⟶ same output (fixed mapping)
- Probabilistic:
Output is probability distribution (used in uncertainty modeling)

(ii):- Boltzmann machines and RBMs
- Boltzmann Machine:
Stochastic network with symmetric connections
- RBM (Restricted Boltzmann Machine):
No connections within same layer
Faster and easier to